The coefficient moments as absolutely convergent series

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Abstract

Completion of Astra #8 rank 4. Two interchanges make the $d = 2$ Taylor-tree coefficients explicit. (1) The face moments: $M[\alpha; \ell; U_{\alpha_i}] = k_1^{-1} \sum_j q_{\gamma_i}(j)\, M[\alpha; \ell; c_{ij}]$, absolutely convergent, by the weighted L^1 interchange under the envelope $|c_{ij}(s)| \leq H(s)\rho^{-i-j}$, $H(s) \leq C_0(1 + s)^D e^{\beta s L}$, with the weights $\sum_j |q_\gamma(j)|\rho^{-j}$ summable. (2) The Gaussian fluctuation moments: for the amplitude array, $c_k(s) = e^{\beta s x_{00}} \sum_{n \leq |k|} \frac{(\beta s)^n}{n!}(y * a^{*n})_k$ is a FINITE sum (degree support), hence $M[\alpha; \ell; c_k] = \sum_{n \leq |k|} \frac{\beta^n}{n!}(y * a^{*n})_k \int_0^\infty s^{\alpha+n-1}(\log s)^\ell e^{-\beta s^2 + \beta s x_{00}} ds$, the paper's moments $\int t^{p/2} e^{-\beta t + \beta\sqrt{t}\,\xi(0)} dt$ in the variable $s = \sqrt{t}$. Zero `sorry`/`axiom`.

1 The things formalised

```
theorem axisPrim_div_summable (  b   :  ) (hb : 0 < b) (hb  : b <  ) :
    Summable fun j :   => |axisPrim   b j| /   ^ j
theorem logMoment_canonU_series (  b   C  L :  ) (h  h  k  k  D :  ) (h  : 0 <  )
    (hb : 0 < b) (hb  : b <  ) (hk  : 0 < k ) (c :   ×   →   →  )
    (hcc :   ij, Continuous (c ij)) (H :   →  )
    (hc :   ij s, |c ij s| *   ^ (ij.1 + ij.2)   H s) (hC  : 0   C )
    (henv :   s, 0   s → H s   C  * (1 + s) ^ D * Real.exp (  * s * L)) (  :  ) (i :  ) :
    (Summable fun j :   => axisPrim ((k  :  ) * uExp h  k  i - h  - 1) b j
       * logMoment   (uExp h  k  i)   (c (i, j)))
      logMoment   (uExp h  k  i)
        (canonU b h  h  k  k  (anaFaceU c b) (fun i j s => c (i, j) s) (uExp h  k  i))
       = 1 / (k  :  ) *  ' j :  , axisPrim ((k  :  ) * uExp h  k  i - h  - 1) b j
           * logMoment   (uExp h  k  i)   (c (i, j))
theorem ampCoeff_eq_sum (  :  ) (x y :   ×   →  ) (k :   ×  ) (s :  ) :
    ampCoeff   x y k s = Real.exp (  * s * x (0, 0))
      *   n   Finset.range (k.1 + k.2 + 1),
          (  * s) ^ n / (n.factorial :  ) * conv y (convPow (dropConst x) n) k
theorem gaussMoment_integrableOn (    x  :  ) (h  : 0 <  ) (h  : 0 <  ) (  n  :  ) :
    IntegrableOn (fun s => s ^ (  - 1) * Real.log s ^
      * (Real.exp (-  * s ^ 2) * (s ^ n * Real.exp (  * s * x )))) (Ioi 0)
theorem logMoment_ampCoeff_eq_sum (    :  ) (h  : 0 <  ) (h  : 0 <  ) (x y :   ×   →  )
    (k :   ×  ) (  :  ) :
    logMoment       (ampCoeff   x y k)
      =   n   Finset.range (k.1 + k.2 + 1),
          ^ n / (n.factorial :  ) * conv y (convPow (dropConst x) n) k
          * logMoment       (fun s => s ^ n * Real.exp (  * s * x (0, 0)))
```

2 Mathematics

For the face moments the term family is $f_j(s) = c_{ij}(s)q_{\gamma_i}(j)$ with $|f_j(s)| \le \frac{H(s)}{\rho^i} \cdot \frac{|q_{\gamma_i}(j)|}{\rho^j}$; the weights are summable because $|q_\gamma(j)| \le b^j b^{-\gamma}/(J-\gamma)$ for $j \ge J > \gamma$ gives a geometric tail $(b/\rho)^j$. So $\sum_j |f_j(s)| \le \frac{C_0 Q}{\rho^i}(1+s)^D e^{\beta s L}$ with $Q = \sum_j |q(j)|\rho^{-j}$, an envelope of the shape accepted by `hasSum_logMoment_of_envelope` (unit 98), and the interchange follows. For the Gaussian moments, $E_m(t) = \sum_{n<M} t^n/n!\,(a^{*n})_m$ for any $M > |m|$ (the extra terms vanish by degree support), so the convolution $(y * E(t))_k$ can be summed over a common range $n \le |k|$ and the two finite sums commute; the finite sum then passes through the s-integral once each term $s^{\alpha-1}(\log s)^\ell e^{-\beta s^2} s^n e^{\beta s x_{00}}$ is integrable, which is the envelope integrability of unit 96 with $R = n$, $a' = x_{00}$.

3 Paper-facing meaning

Combining units 121–126, every coefficient of the $d = 2$ Taylor tree (equal starts) is now an explicit absolutely convergent series in the convolution data $(y * a^{*n})_{ij}$ of the coefficient arrays of η and $\xi - \xi(0)$, weighted by $b^{j-\gamma}/(j-\gamma)$ (or $\log b$) and by the Gaussian moments $\int_0^\infty s^{\alpha+n-1}(\log s)^\ell e^{-\beta s^2+\beta s\xi(0)}ds$. This is the content of the paper's "coefficients are absolutely convergent series expressed as derivatives of ξ, η and the fluctuation function" once the coefficient arrays are identified with Taylor data (the 1D identification exists; the 2D one is Taylor-coefficient uniqueness for double series). What remains for the paper's $d = 2$ theorem is the unequal-starting-exponent case (Astra #8 rank 3), which needs a new corner formula and an audit of the face expansion for negative γ.

4 Lean notes

- `HasSum.congr_fun` wants the equation in the direction "new term = old term".
- `Finset.range_subset` is no longer an iff usable with `.2`; use `Finset.range_mono`.
- `Finset.sum_comm` needs the inner sum exposed: push the outer factor in with `Finset.mul_sum` inside the `sum_congr` first.
- `gcongr` may leave a single arithmetic side goal (here $s \le 1 + s$) rather than one per factor; bullets then fail with "no goals".
- `positivity` cannot see $0 \le s^{\alpha-1}$ from $s \in \mathrm{Ioi}\, 0$; pass `Real.rpow_nonneg hs.le _`.
- `integral_finset_sum` is deprecated for `integral_finsetSum`.
- Writing `div_eq_mul_inv b ^ j` parses as `(div_eq_mul_inv b) ^ j`; prove the rpow identity as a separate `have`.